Ideation Phase
Brainstorm & Idea Prioritization Template

Date	9 March 2026
Project Name	Malaria Detection using Deep Learning
Maximum Marks	4 Marks

Step-1: Team Gathering, Collaboration and Problem Statement

Malaria remains one of the most serious infectious diseases in many parts of the world, especially in developing countries. Early detection of malaria is extremely important because delayed diagnosis can lead to severe health complications and even death. Traditionally, malaria diagnosis is done by manually examining blood smear images under a microscope. This process requires skilled laboratory technicians and can be time-consuming, expensive, and prone to human error.

With the advancement of Artificial Intelligence and Deep Learning, automated medical image analysis has become a promising solution for disease detection. Deep learning models such as Convolutional Neural Networks (CNNs) can analyse microscopic images of blood cells and classify whether the cells are infected with malaria parasites or not.

The main problem addressed in this project is the **lack of fast, accurate, and automated malaria diagnosis systems**, especially in regions with limited medical resources. To solve this issue, our team proposes a **deep learning-based malaria detection system** that can automatically classify blood smear images into infected and uninfected categories. The system will use trained deep learning models and a web interface that allows users to upload blood smear images and obtain instant predictions.

This solution can help healthcare professionals by providing a faster screening tool and reducing dependency on manual diagnosis.

Step-2: Brainstorm, Idea Listing and Grouping

During the brainstorming session, the team discussed various ideas related to healthcare diagnostics and artificial intelligence. The following ideas were proposed:

Idea List

1. **AI-based Malaria Detection System**
   Use deep learning models to classify blood smear images as infected or uninfected.
2. **Mobile Application for Malaria Detection**
   A smartphone application that allows users to capture microscope images and detect malaria instantly.
3. **Cloud-based Medical Diagnosis Platform**
   A centralized system where hospitals can upload blood sample images for AI-based analysis.

4. **Automated Laboratory Assistance Tool**
   Software that assists laboratory technicians by highlighting infected cells automatically.
5. **Dataset Enhancement using Image Augmentation**
   Improve model accuracy by applying data augmentation techniques to medical datasets.
6. **Lightweight Deep Learning Model for Rural Clinics**
   Develop a lightweight model that can run on low-power systems in rural healthcare centers.
7. **Multi-Disease Detection System**
   Extend the model to detect other blood-related diseases such as anaemia or dengue.

After evaluating these ideas, the team grouped them into the following categories:

Group 1 – Disease Detection using AI

- AI-based Malaria Detection System
- Multi-disease Detection System

Group 2 – Healthcare Tools

- Automated Laboratory Assistance Tool
- Cloud-based Medical Diagnosis Platform

Group 3 – Accessibility Solutions

- Mobile Application for Malaria Detection
- Lightweight Model for Rural Clinics

<u>Step-3: Idea Prioritization</u>

To select the most suitable idea, the team evaluated each concept based on two main factors:

- **Importance** – How valuable the solution is for healthcare and society.
- **Feasibility** – How practical it is to implement using available resources and time.

After analysing all the ideas, the team concluded that **AI-based Malaria Detection using Deep Learning** has both high importance and high feasibility.

Reasons for Selecting the Final Idea

1. Malaria is a major global health problem and requires faster diagnosis methods.
2. Deep learning models are highly effective for image classification tasks.
3. The availability of malaria blood smear datasets allows efficient training of models.
4. The system can be implemented as a web application for easy accessibility.
5. The project can significantly assist healthcare professionals by reducing manual workload and improving diagnostic speed.

Therefore, the team finalized **Malaria Detection using Deep Learning** as the primary project idea.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Our Problem Statement

Lack of fast, accurate, and automated malaria diagnosis systems in regions with limited medical resources.

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Step-2: Brainstorm, Idea Listing and Grouping

Idea Groups

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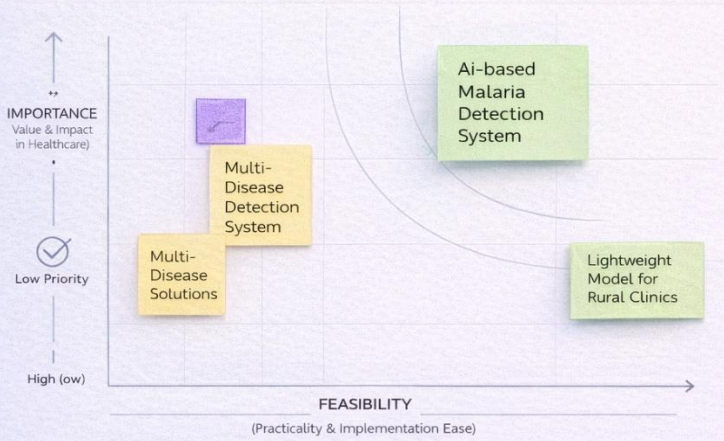

Why we selected AI-based Malaria Detection System

- Malaria is a major global health problem and requires faster diagnosis methods.
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